Collision-Graph Schur Envelope for the Literal Prime Shell

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August 2026

Abstract

TPC-220 rewrites the q-transverse term of the literal prime shell as a multiplicative collision Gram, but leaves open how much an absolute collision estimate can save. This paper supplies the next exact layer. The row Gram is positive semidefinite, its weighted shell energy has an exact quadratic-form representation, and a weighted Schur test gives a reusable operator envelope in terms of collision degrees. We then construct a literal finite saturation: with $h = 5$, $H = 500$, constant profile, and primes $101, 151, 181, 191$, every row is $e_1 + e_4$. The Gram is therefore $2J_4$ and the coherent-to-diagonal ratio is exactly $P = 4$. Thus absolute Schur control is a valid structural envelope but cannot, by itself, yield a sub-P saving. No prime-distribution estimate or arithmetic L^2 advance is claimed.

Claim level. `PROVED_STRUCTURAL_L1 / COLLISION_GRAPH_SCHUR_ENVELOPE`.

1 The collision object

TPC-219 identified the exact condition for improving the scalar q-shell collapse: transverse q-energy must be large. TPC-220 put that condition back in arithmetic coordinates. We continue with the same literal rows, but suppress the packet superscript for readability. Fix an active denominator h, a prime q coprime to h, and write

$$L_{h,q} = \left\lfloor \frac{hq}{H} \right\rfloor, \qquad \mathcal{M}^*_{h,q} = \{m \in \mathbb{Z} : 0 < |m| \le L_{h,q},\ (m,h) = 1\}.$$

For a profile ψ, the row on primitive residues is

$$B_{h,q}(a) = \sum_{m \in \mathcal{M}^*_{h,q}} \psi\left(\frac{Hm}{hq}\right) \mathbf{1}_{mq^{-1} \equiv a \ (h)}. \tag{1}$$

Let $\mathcal{Q}$ be a finite active prime set and regard B_q as a vector indexed by $a \pmod h$ with $(a,h) = 1$. Define

$$\Gamma_h(q,q') = \sum_{\substack{a \pmod h \\ (a,h)=1}} B_{h,q}(a)\overline{B_{h,q'}(a)}. \tag{2}$$

The TPC-220 crosswalk gives the literal expansion

$$\Gamma_h(q,q') = \sum_{m \in \mathcal{M}^*_{h,q}} \sum_{m' \in \mathcal{M}^*_{h,q'}} w_{h,m,q}\overline{w_{h,m',q'}}\mathbf{1}_{mq' \equiv m'q \ (h)}, \tag{3}$$

where $w_{h,m,q} = \psi(Hm/(hq))$. Thus the matrix is not an abstract covariance surrogate: its off-diagonal entries are precisely the multiplicative collision edges.

2 Exact PSD energy and Schur control

Theorem 1 (PSD identity)**.** *The matrix* $\Gamma_h = (\Gamma_h(q,q'))_{q,q'\in\mathcal{Q}}$ *is Hermitian positive semidefinite. For every complex weight vector* $\lambda = (\lambda_q)_{q\in\mathcal{Q}}$,

$$\sum_{\substack{a \pmod h\\(a,h)=1}} \left|\sum_{q\in\mathcal{Q}} \lambda_q B_{h,q}(a)\right|^2 = \lambda^*\Gamma_h\lambda. \tag{4}$$

Proof. Let A be the matrix whose row indexed by q is B_q. Then the definition (2) is $\Gamma_h = AA^*$, up to the harmless choice of whether vectors are written as rows or columns. Expanding the left side of (4) gives the stated quadratic form. A matrix of the form AA^* is Hermitian positive semidefinite. □

Theorem 2 (Weighted Schur envelope)**.** *Let* $p_q > 0$ *be arbitrary. Then*

$$\lambda^*\Gamma_h\lambda \le \rho_h(p)\,\|\lambda\|_2^2, \qquad \rho_h(p) = \max_{q\in\mathcal{Q}} \frac{1}{p_q}\sum_{q'\in\mathcal{Q}} |\Gamma_h(q,q')|p_{q'}. \tag{5}$$

In particular, for $p_q = 1$,

$$\lambda^*\Gamma_h\lambda \le \left(\max_q \sum_{q'} |\Gamma_h(q,q')|\right)\|\lambda\|_2^2. \tag{6}$$

Proof. By (4), take absolute values termwise and use

$$2|\lambda_q\lambda_{q'}| \le \frac{p_{q'}}{p_q}|\lambda_q|^2 + \frac{p_q}{p_{q'}}|\lambda_{q'}|^2.$$

After summing over the two orientations of (q,q'), the coefficient of $|\lambda_q|^2$ is at most

$$p_q^{-1}\sum_{q'} |\Gamma_h(q,q')|p_{q'}.$$

Taking the maximum proves (5), and (6) is the choice $p_q = 1$. □

The quantity in parentheses in (6) is the absolute collision degree. Formula (5) is useful when the q-family has an inhomogeneous cutoff or when a prospective estimate naturally assigns different weights to different prime labels. Neither formula introduces a sign or phase cancellation: all off-diagonal entries are replaced by their absolute values.

3 A literal saturation of the absolute envelope

Theorem 3 (Finite aligned obstruction)**.** *Take* $h = 5$, $H = 500$, *the constant profile* $\psi \equiv 1$, *and*

$$\mathcal{Q}_{\mathrm{sat}} = \{101, 151, 181, 191\}.$$

Then $L_{h,q} = 1$ *for every* $q \in \mathcal{Q}_{\mathrm{sat}}$ *and*

$$B_{h,q} = e_1 + e_4, \qquad \Gamma_h = 2J_4.$$

Consequently the Schur radius and top Rayleigh quotient are both 8*, while equal weights have energy* 32 *and the total diagonal is* 8*. Their ratio is exactly* $4 = |\mathcal{Q}_{\mathrm{sat}}|$.

object	scope	status
PSD/collision Gram	3 moduli, 2 profiles	exact
Schur envelope	6 rational records	exact
weighted Schur	positive test weights	exact
literal saturation	$h = 5$, 4 primes	ratio $P = 4$

Table 1: TPC-221 certificate scope and claim boundary.

Proof. All four displayed numbers are primes congruent to 1 (mod 5). Moreover $\lfloor 5q/500 \rfloor = 1$, so the only nonzero atoms are $m = 1$ and $m = -1$. Since $q^{-1} = 1 \pmod 5$, both atoms land at the primitive residues 1 and 4. Thus every row is $e_1 + e_4$. The inner product of any two rows is 2, so $\Gamma_h = 2J_4$. Its row sum and its top eigenvalue are 8. For the all-one weight vector, the two occupied coordinates each have amplitude 4, hence the energy is $2 \cdot 4^2 = 32$. The four diagonal entries sum to 8, giving the ratio 4. □

This is a literal finite saturation, not an arbitrary PSD counterexample. It proves a scoped methodological obstruction: a theorem that only bounds absolute collision degrees cannot guarantee a saving below the q-cardinality factor uniformly over the literal class. It does not say that the same alignment survives in a growing prime window.

4 Certificate and route position

The release certificate checks (3)–(6) on the TPC-220 rational fixtures, replays the weighted test with an independent implementation, and separately checks the saturated fixture. The exact finite summary is:

The strongest positive result is a portable operator envelope for the exact collision graph. The strongest obstruction is the literal saturation above. The next theorem must use signed or phase-sensitive structure of the off-diagonal graph; an absolute Schur argument cannot pay the missing transverse lower bound.

`ARITHMETIC_ADVANCE=NO,    L2=NONE,    FULL_GATE_B=OPEN.`

5 Conclusion

TPC-221 turns the TPC-220 collision object into a precise PSD/Schur interface and closes a natural false shortcut. The envelope is proved and reusable, but its literal aligned saturation shows why the next step must control signs, phases, or genuine dispersion rather than merely count collision edges. Twin-prime reassembly remains open.

References

[1] TPC-220 repository proof record, “Prime-AP collision crosswalk for the literal prime shell,” 2026. Internal source lock; no external asymptotic theorem is invoked.